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Vibrations of Anarchy

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Abstract

This report presents *Vibrations of Anarchy*, a browser-based AR project developed from the poetry and political legacy of Joaquim Moreira da Silva, also known as Poeta Carpinteiro. The project uses image tracking, 3D animation, sound, and short textual fragments to create an augmented reality scene in which a virtual hammer strikes an open book and gradually reveals phrases taken from the poet’s writings. The work connects carpentry, labour, literacy, and political expression through a simple interactive structure. The report describes the conceptual basis of the project, its visual and artistic decisions, the technological tools used in development, and the main technical challenges encountered during implementation. The project explores augmented reality as a format for presenting literary and historical material through action rather than static display.

Keywords

Augmented Reality; WebAR; interactive poetry; digital archive; literary heritage

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Introduction

Augmented reality is often used to overlay digital content on physical objects. In literary and archival contexts, this creates a question of how AR can be used not only to display information, but to activate text through interaction.

This project examines that question through the life and writings of Joaquim Moreira da Silva. It focuses on the relation between labour, poetry, and political expression, and uses AR to test how book, tool, and phrase can function together as an interactive system.

Theory and State of the Art of Augmented Reality

Augmented Reality connects physical objects with digital text, sound, and animation. In book-based AR, printed pages are expanded with 3D elements and interactive triggers, creating a hybrid form between paper and screen. Research shows that such formats can increase behavioral, cognitive, affective, and social engagement (Alhamad et al., 2024).

This is relevant to Vibrations of Anarchy, where AR is used not only to display text, but to connect the viewer with the poet’s political and social ideas. In AR art practice, digital layers can turn a static object into an interactive experience and make the audience more active (Adeyemi, 2025).

An important idea in AR theory is the use of conceptual metaphors. One of them is the active print metaphor, in which digital content is attached to a printed surface and appears through interaction. This is directly relevant to Vibrations of Anarchy, where the book functions as the surface that reveals hidden phrases (Fomina, 2023).

Another important concept is reality-based interaction, where digital actions follow familiar real-world behaviors. Research shows that this often feels more intuitive than abstract command-based systems. In Vibrations of Anarchy, the hammer follows this principle: the user clicks, the hammer strikes the page, and the text appears (Franzluebbers et al., 2020).

A useful recent example is PropType, which turns everyday objects, including books, into interactive AR surfaces. This shows how physical objects can serve as tactile anchors for digital interaction. In Vibrations of Anarchy, the book works in a similar way, not only as a marker, but as a material object that supports interaction and meaning(Gil et al., 2025).

Overall, current AR theory and practice show that AR experiences often depend on a physical object, intuitive interaction, and a clear concept. Vibrations of Anarchy follows this structure through the book as an active printed surface, the hammer as a reality-based tool, and the slogans as a digital layer revealed through action.

References and Inspirations

Vibrations of Anarchy draws on AR and digital art works where the book functions as an interface and text appears through interaction. These references informed the use of the open book as a target image, the empty page as a starting surface, and language as something revealed rather than already present.

Andromeda by Caitlin Fisher uses a pop-up book with AR markers that activate digital text, video, and sound through the camera. This was relevant because it shows how a physical book can generate poetic content through AR interaction (Fisher, 2008).

Between Page and Screen by Amaranth Borsuk and Brad Bouse uses printed markers instead of visible text. The poem appears on screen only when the page is recognized by a webcam. This supported the idea of the page as a surface that reveals hidden language through interaction (Borsuk & Bouse, 2012).

Voidopolis by Kat Mustatea presents damaged pages that can only be read through an AR app, while the digital text changes over time. This was relevant because it treats text as unstable and dependent on the relation between page and digital (Mustatea, 2023).

The end-credit sequence of Sherlock Holmes by Prologue was used as a visual reference. It shows ink, engraving, and printed fragments forming images and text. This informed the idea of phrases appearing as traces produced through force, impact, and printed matter (Prologue Films, 2009).

Together, these references informed the main structure of the project: the book as target and archive, the page as a surface of activation, and the phrases as short textual fragments that appear through action.

Design and Artistic Options and Decisions

I based the project on the life and writings of Joaquim Moreira da Silva as a carpenter, worker-poet, and anarchist voice. In his history, manual labour, political struggle, and language are directly connected: he entered the labour and anarchist milieu in Porto, his poetry became a vehicle for social and political ideas, and he later taught literacy, treating reading and writing as tools of action. His texts also circulated through pamphlets and small printed materials.

This led to the final concept: Vibrations of Anarchy. The user points the camera at an image of an open empty book. A 3D hammer appears above the pages. Each click makes the hammer strike the book, and short slogan-like phrases taken from the poet's writings appear inside the spread.

This structure was important for three reasons. First, the book functions as both an archive and a place of memory. It refers to his manuscripts, notebooks, and collected writings. Second, the hammer introduces the physical dimension of labour. Although it is a simplified symbolic choice, it creates a direct link to carpentry, tools, and work. Third, the appearing phrases refer to the leaflets and printed political texts through which he spread his ideas. In this way, the book is not presented as passive storage, but as a place where language can be struck, released, and returned to public force.

The title Vibrations of Anarchy comes directly from the language of his poetry and became a key conceptual anchor. It expresses the idea that poetry is not static text but a form of resonance, agitation, and living energy. This title helped unify the visual and interaction decisions: the appearing phrases are not just textual content, but vibrations released by impact, memory, and ideological force.

Figure 1. Development of the target image, visual references, and final AR scene composition

I studied the visual material of his books and manuscripts, especially A Lira do Povo and Minha Vida e Minhas Obras, and used them as the basis for the scene. This process is shown in Figure 1, where the upper row presents the source visual material and the first phrase-placement tests, and the lower row shows the development of the final target image and interactive composition. From the book photographs, I built a blank open spread to serve as the main image target. I experimented with several target images, but a plain old page was difficult to track reliably.

To solve this, I added the ornamental border from the title page. This step can also be seen in Figure 1, where the plain spread is later transformed into a framed target image. This also became an artistic choice: the ornament is simple, naive, and floral, and it gives the target stronger visual identity while keeping a connection to the original book.

For the interactive object, I chose a hammer rather than a neutral cursor. The hammer creates a direct link to carpentry and labour, and it makes the phrases appear through impact rather than passive display. In Figure 1, the hammer is shown as part of the target composition in the lower middle image, where it becomes the main interactive tool within the scene. I used a 3D hammer model (Tics, n.d.).

For the textual elements, I selected short phrase fragments from the available manuscript pages rather than full poems. This made the interaction clearer and closer to the form of a leaflet, slogan, or political call. The phrases were placed so that they would emerge from the page after each strike. This development is also visible in Figure 1, where the phrase fragments first appear as dispersed tests on the open spread and later as more structured slogan-like elements in the final composition.

For sound, I added a hammer impact effect and a lyre-harp track. The hammer sound reinforces the gesture of labour and activation. The lyre track connects the scene to the poet's repeated image of the lira as a political and poetic instrument. I used a hammer sound effect (Universfield, 2025) and a lyre-harp track (PJCreations, 2026).

Overall, the design is based on one main decision: to present poetry not as static text, but as something released through labour, impact, and historical memory.

Development Report

Tools, Languages & Platforms

The project was built with web technologies. HTML5 defined the page structure, and JavaScript handled interactivity and animation. No CSS framework was used beyond basic reset styles.

The AR layer was built with MindAR.js (v1.2.5), a browser-based image tracking library. 3D rendering used A-Frame (v1.4.0), built on top of Three.js. The hammer model was used as a GLTF file with its geometry and texture maps.

For local development, Node.js served the files through npx serve. Because camera access required HTTPS, ngrok exposed the local server to mobile browsers during testing. Blender was used to inspect and attempt rotation of the 3D model. The MindAR Image Targets Compiler was used to process target images and export .mind files.

Development Process

The project first used 8th Wall. The aframe-image-targets-example repository was cloned and set up locally, which required fixing webpack errors, including a binary .DS_Store file being processed as JavaScript and missing .json target files. After these fixes, the experience still failed to load because the 8th Wall hosted platform had retired its service in February 2026.

A second attempt used the 8th Wall Desktop application, but the preview window never loaded and remained on “Initializing...”. After investigation, 8th Wall was abandoned and replaced with MindAR.js.

The core AR loop was then built in MindAR step by step: target compilation, model loading, animation, and phrase reveal were added and tested separately on a physical iPhone through ngrok.

Challenges & Errors

Image target quality was the main challenge. The book pages were almost blank and produced no recognition points in the MindAR compiler. Different photos were tested. The solution was to add a decorative hand-painted border, which provided enough visual features for tracking. No text was required.

3D model orientation also required repeated testing. Because GLTF models use their own coordinate system, the hammer appeared flat or incorrectly rotated in AR space. Attempts to fix this in Blender failed because Apply Transforms (Ctrl+A) did not persist correctly on export. The final result was achieved by manually setting the A-Frame rotation attribute to rotation="90 180 -135".

Audio autoplay was blocked until the first user gesture, so music playback was started inside the first click event. ngrok port changes sometimes caused ERR_NGROK_8012, which required restarting the ngrok tunnel. Target recognition also failed after image changes because the browser cached the old .mind file, so the mobile browser had to be hard-refreshed or its site data cleared.

Result

The result is a WebAR experience that runs in a modern mobile browser without app installation. The user points the camera at the book, a 3D hammer appears, and each tap reveals a new phrase until all eight are placed on the pages. Music plays while the book is tracked and pauses when it leaves the frame.

Discussion and Conclusion

The project uses AR to connect literary material with gesture, object, and surface. In this case, the poem is not accessed through continuous reading, but through repeated action. This changes the role of the text: it appears in fragments, through impact, and in direct relation to the book as a physical support. Because of this, the work shifts the poem from a fixed literary form toward an activated public statement.

This structure also changes the role of the book. It is not used only as a document or historical reference, but as a site where writing can reappear. The phrases do not function as illustration or caption. They function as extracted fragments of political language, closer to slogans, pamphlets, or calls. The hammer is part of this logic. It introduces labour not as background context, but as the condition through which language is released.

From this, the project suggests a specific use of AR for literary and historical material. Instead of reconstructing the past or adding information on top of an object, AR can be used to stage a relation between archive, action, and appearance. In Vibrations of Anarchy, the digital layer does not replace the book. It makes the book operate differently: as archive, trigger, and surface of reactivation.

Future development could strengthen this relation in visual, sonic, and narrative ways. The text reveal could become more material, for example through small wood shavings that emerge at the moment of impact, turn into ink drops, and then settle into letters. The sound design could become more layered, with the lyre growing clearer and stronger after each strike, so that repeated work gradually produces resonance. The text animation could also be refined, with letters vibrating at the moment they appear, as if responding to a real physical удар. A larger version of the project could also expand from one page to several pages, allowing the interaction to unfold as a sequence: craft, literacy, writing, political speech, and the circulation of ideas.

In conclusion, the project places poetry, labour, and political memory into one interactive system. It proposes AR as a medium for presenting text not only as content to read, but as something that can be struck, released, and returned to view through action.

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